Voltage Source and Current Source

Defn: A voltage source is a device which can produce a continuous force to move electrons (or continuous voltage) through the wire connected into the two terminals of the device

Types of voltage source

1. Direct voltage source

It is a device which produces a continuous direct voltage output

A direct voltage is a type of voltage whose polarity remains the same. This type of voltage causes the current to move only in one direction continuously.

Examples of direct voltage source: Cells, Battery, DC generator.

* Polarity is defined as having two opposite electrical charges. Good and evil are an example of polarity.

2. Alternating voltage source

It is a device which produces an

alternating direct voltage output.

Example: AC Generator, DC to AC converter, etc.

An alternating voltage is the kind of voltage whose polarity is reversed periodically.

Alternating voltage causes the current to move in one direction for a period and then in another direction for another period.

Ideal Voltage Source

It is a kind of voltage source whose internal resistance is zero such that the supplied voltage does not change even if the external load resistance is changed.

CURRENT SOURCE

It is a device which ~~produces~~ provides the regular flow of electrons or current on a circuit.

(2)

It is a type of voltage source which has enough EMF and surplus electrons so as to produce the flow of electrons.

DIRECT CURRENT SOURCE

It is a current source made of a direct voltage source

Alternatif Current source

It is the type made of an alternatif voltage source.

IDEAL CURRENT SOURCE

It is a current source which provides a constant current without any relation with the voltage supplied to the load.

*

CURRENT ~~AND~~, VOLTAGE, RESISTANCE AND OHM'S LAW

Current is the rate at which electric charge flows past a point in a circuit. In other words, it is the rate of flow of electric charge.

~~$I = Q/$~~

$$\text{Current } (I) = \frac{\text{Electric charge}}{\text{Time}} = \frac{Q}{T}$$

SI Unit: Ampere (A) or amps or amperage

1 ampere = 1 coulomb / second

Voltage also called electromotive force is the potential difference in charge between two points in an electrical field. In other words, voltage is the energy per unit charge.

SI Unit: Volt = 1 joule / coulomb

Relationship

Current is the effect (voltage being the cause). Current can not flow without voltage.

Voltage can exist without current.

Comparison

Current	Voltage
1. It is the same through all components connected in series	voltage gets distributed over components connected in series
2. It gets distributed over components connected in parallel	voltages are the same across all components connected in parallel.

RESISTANCE :- This is a property of conductor by virtue of which the passage

of current is opposed, causing electric energy to be transformed.

OHM'S LAW

This states that the current through a conductor between two points is directly proportional to the voltage across the two points

$$V \propto I$$
$$V = IR$$

The constant of proportionality is the resistance.

Current — amperes
Voltage — volts
Resistance — Ohms.

KIRCHOFF'S CURRENT LAW

This law states that the algebraic sum of currents in a junction (or node) is zero.

CIRCUIT

A voltage source has two points that have a difference in electrical potential. When there is a closed loop path between these two points, it is called a circuit and current can flow. In the absence of a circuit, current will not flow even if there is voltage.

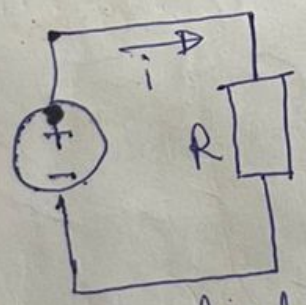


Fig: An electric circuit with a voltage source (e.g. a battery) and a resistor.

Circuit Laws

THEVENIN'S THEOREM

It states that it is possible to simplify any linear circuit no matter how complex to an equivalent circuit with just a single voltage source and series resistance connected to a load.

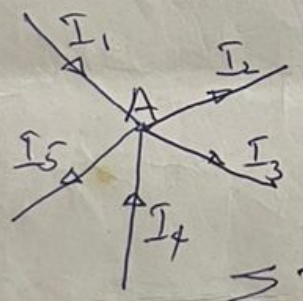
NORTON'S THEOREM

This states that it is possible to simplify any linear circuit, no matter how complex, to an equivalent circuit with just a single current source and parallel resistance connected to a load.

KIRCHHOFF'S LAWS

Kirchhoff's Current Law (KCL)

It states that in any electrical network, the algebraic sum of the current meeting at a point (or junction) is zero. It simply means that the total current leaving a junction is equal to the total current entering that junction.



$$I_1 + I_4 - I_2 - I_3 - I_5 = 0$$

$$I_1 + I_4 = I_2 + I_3 + I_5$$

$$\Sigma I = 0$$

2. KIRCHHOFF'S VOLTAGE LAW

It states that: The algebraic sum of the products of currents and resistances in each of the conductors in any closed path in a network plus the algebraic sum of the emfs in that path is zero.

$$\Sigma IR + \Sigma emf = 0$$

SUPERPOSITION THEOREM

theorem states that:

a network of linear resistances containing more than one generator (or source of e.m.f), the current which flows at any point is the sum of all the currents which would flow at that point if each generator were considered separately and all the other generators replaced for the time being by resistances equal to their internal resistances.

Superposition Theorem

In a network of linear resistances containing more than one generator (or source of emf), the current which flows at any point is the sum of all the currents which would flow at that point if each generator were considered separately and all the other generators replaced for the time being by resistances equal to their internal resistances.

or The current in a branch of a linear circuit is equal to the sum of the currents produced by each source with the other sources set equal to zero.

Thevenin Theorem

The current flowing through a load resistance R_L connected across any two terminals A and B of a linear, active bilateral network is given

The current in any load of a linear network is the same as if the load were connected to a single generator whose emf is equal to the open circuit voltage

across the two terminals and shorting internal resistance is the resistance of the network looking back into these terminals with all the generators replaced by their internal resistances.

*The theorem can be more simply stated as: "Any linear, two-terminal d.c. network can be represented by a voltage source in series with a resistance".

Gain – ability of a tiny input signal to be increased in size of a signal the for output

(A)

1. INTEGRATED CIRCUITS (IC)

An IC is a complex electronic circuit in which both the active and passive components are fabricated on a tiny single chip of silicon

Active components are those which have the ability to produce gain. Examples are transistors and Field effect Transistors (FETs)

Passive components or devices are those which do not have this ability. Examples are resistors, capacitors and inductors

Advantages of ICs

1. extremely small physical size

Often the size is thousands of times smaller than a discrete circuit. The various components and their interconnections are distinguishable only under a powerful microscope. Note: a discrete circuit is one that is built by connecting separate components. Each component is produced separately and then all

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work for longer periods without aging

~~2. Very small weight~~
are assembled together to make the electronic circuit

2. Very small weight
Since many circuit functions can be packed into a small space, complex electronic equipment can be employed in many applications where weight and space are critical such as in aircraft or space-vehicles.

3. Reduced Cost
The reduction in cost per unit is due to the fact that many identical circuits can be built simultaneously on a single wafer. this process is called batch fabrication.

4. Extremely high reliability
This is due to many factors
(a) absence of soldered connections
(b) need for fewer interconnections which is major cause of circuit failures.
(c) small temperature rise due to low power consumptions of ICs
* Higher reliability means that ICs will

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(8)

work for longer periods without giving any trouble.

5. Increased response time and speed

Since various components of an IC are located close to each other in or on a silicon wafer, the time delay of signals is reduced.

6. Low Power Consumption

Because of their small size, ICs are more suitable for low power operation than bulky discrete circuits

7. Easy Replacement

ICs are hardly ever repaired because in case of failure, it is more economical to replace them ~~that~~ than to repair them.

8. Higher yield

The yield is the percentage of usable devices. Because of the batch fabrication, the yield is very high. Faulty devices usually occur because of some defect in

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(c)

2. MSI – Medium Scale Integration

the silicon wafer or in the fabrication steps.

DRAWBACKS OF ICS

The integrated circuits suffer from the following drawbacks:

1. Coils or inductors can not be fabricated
2. ICs function at fairly low voltages
3. They handle only limited amount of power
4. They are quite delicate and can not withstand rough handling or excessive heat.

SCALE OF INTEGRATION

An approximate method of classifying the amount of circuit or component density is as follows:

1. SSI – Small scale Integration
The number of circuits contained in one IC package is less than 12.

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Because [illegible] more [illegible] than bulky discrete [illegible]

7. Easy Replacement
ICs are hardly ever repaired [illegible] in case of failure, it is more [illegible] to replace them than to [illegible]

8. Higher yield
The yield is the percentage [illegible] devices. Because of the batch [illegible] The yield is very high. For [illegible] usually occur because of [illegible]

(c)

2. MSI – Medium Scale Integration
Number of circuits per IC package is between 13 and 99.

3. LSI – Large Scale Integration
The number of circuits per IC package is between 100 and 9,999.

4. VLSI – Very Large Scale Integration
Number of circuits per package is between 10,000 and 99,999.

5. ULSI – Ultra Large Scale Integration
The circuit density is between 100,000 and 999,999.

6. GSI – Giga Scale Integration
The number circuits per package is 1,000,000 or more

CLASSIFICATION OF ICs BY STRUCTURE

Structurally speaking ICs can be classified into the following three types:

1. Monolithic Integrated Circuits
In this type IC, all circuit components (both active and passive) are fabricated

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ICs ...
in case of failure ...
to replace them ...
8. Higher yield
The yield is the percentage ...
devices. Because of the batch ...
The yield is very high. Fa...
usually occur because of ...

inseparably within a single continuous piece of silicon crystalline material called wafer (or substrate). All components are automatically part of ~~one~~ the same chip.

2. Thick and Thin-Film ICs

Thin film ICs are constructed by depositing films (typically 0.1 to 0.5 μm) of conducting material through a mask on the surface of substrate made of glass or ceramic.

Thick film ICs :- Silk screen printing techniques are used to create desired circuit patterns on the surface of the substrate.

3. Hybrid or Multichip ICs

These are formed either by inter-connecting a number of individual chips or by a combination of film and monolithic IC techniques.

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6. [illegible] the number [illegible] or more (1,000,000)

Classification of ICs by structural operation

ICs can be classified into three types:

1. Monolithic Integrated circuit
   - this type IC (all circuit both active and passive) are

They are frequently used in

1. Operational amplifiers
2. Small-signal amplifiers
3. power amplifiers
4. RF and IF amplifiers
5. Microwave Amplifiers
6. Multiplier:
7. Voltage Comparators
8. Voltage Regulators

DIGITAL INTEGRATED CIRCUITS

These are mostly used by the computer industry. They lend themselves easily to monolithic integration because a computer uses a large number of identical circuits.

Digital ICs contain circuits whose input and output voltages are limited to two possible levels — low or high. It is because digital signals are usually binary.

Digital ICs include circuits such as

1. logic gates (2) flip-flops (3) counters (4) clock chips (5) Calculator chips (6) Memory chips (7) Microprocessors (MP) etc.

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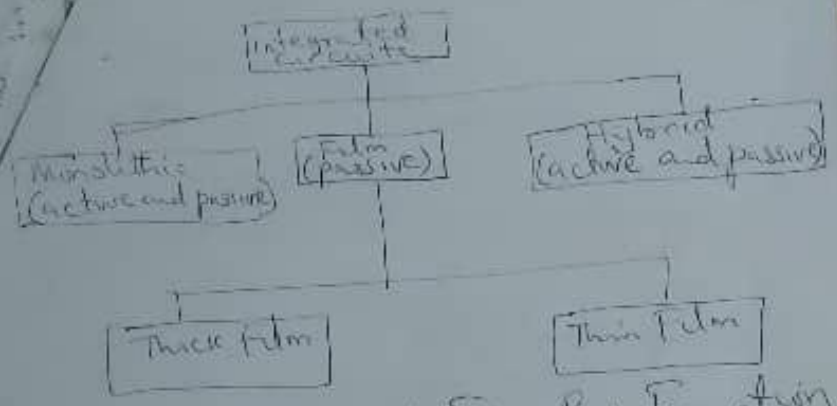
1,000,000 or more

CLASSIFICATION OF ICs BY

Structurally speaking ICs
into the following three ty
1. Monolithic Integrated Cir
- this type IC, all cir
both active and passive) o

(D)



Classification of ICs By Function

1. Linear
2. Digital

(1). Linear Integrated Circuits (LICs)

LICs are also referred to as analogue ICs because their inputs and outputs can take on a continuous range of values and the outputs are generally proportional to the inputs. As compared to digital ICs

(a) LICs are used much less
(b) They possess much higher reliability because so many external connections are are eliminated.

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(E)

IC TERMINOLOGIES

1. Bonding – attaching the die on the ceramic substrate and then connecting the leads to the package
2. Chip – an extremely small part of a silicon wafer on which IC is fabricated
3. Circuit Probing – testing the electrical performance of each IC chip ~~the~~ with the help of a microscope and multi-point probe.
4. Die – same as a chip
5. Diffusion – a process that consists of the introduction of impurities into selected regions of a wafer to form junctions
6. Encapsulation – putting a cap over the IC and sealing it in an inert atmosphere
7. Epitaxy – a process of the controlled growth of a crystalline doped layer of silicon on a single crystal substrate
8. Etching – a process of selective removal of regions of a semiconductor metal or silicon dioxide

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Semiconductor 70123 Pg 2030 (A)

A semiconductor ...

9. Mask – a glass plate with desired patterns for diffusion or metallization. Usually a single mask is not ~~enough~~ to sufficient to fabricate a IC.

10. Metallization – a process for providing ohmic contacts and interconnections by evaporating aluminium over the chip.

11. Photolithography – a process to transfer geometrical pattern from the mask to the surface of the wafer

12. Photoresist – a light-sensitive material that hardens when exposed to ultraviolet light

13. Wafer – a thin disk of semiconductor in which number of ICs are fabricated simultaneously.

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(a) Intrinsic Semiconductors

An intrinsic semiconductor is one which is made of the semiconductor material in its ~~exp~~ extremely pure form. [Alternatively, an intrinsic semiconductor may be defined as one in which the number of conduction electrons is equal to the number of holes.]

Common examples of such semiconductors are: pure germanium and silicon which have forbidden energy gaps of 0.72 eV and 1.1 eV respectively. The energy gap is so small that even at ordinary room temperature, there are many electrons which possess sufficient energy to jump across the energy gap between the valence and the semiconduction bands. However it is worth noting that for each electron liberated into the conduction band, a positively charged hole is created in the valence band.

(b) Extrinsic Semiconductors

Those intrinsic semiconductors to which some suitable impurity or doping agent or dopant has been added in extremely small amounts (about 1 part in 10^8) are called extrinsic or impurity semiconductors

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Pg 2080 (A)

SEMICONDUCTORS

A semiconductor material is one whose electrical properties lie between those of insulators and conductors e.g germanium and silicon

In terms of energy bands, semiconductors can be defined as those materials which have almost an empty conduction band and almost filled valence band with a very narrow energy gap separating the two.

Semiconductor current is the sum of electron and hole currents flowing in opposite directions.

Types of semiconductors

Semiconductors can be classified as shown below:

- Semiconductors
  - Intrinsic or pure Semiconductors
  - Extrinsic or impure Semiconductors
    - N-type
    - P-Type

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Types of semiconductors
Intrinsic or pure semiconductors
Extrinsic semico
N-type

(5)

The usual doping agents are
1. Pentavalent atoms having five valence electrons (arsenic, antimony, phosphorous) or
2. trivalent atoms having three valence electrons (gallium, indium, aluminium, boron).

Depending on the type of doping material used, extrinsic semiconductors can be subdivided into two classes
(i) N-type semiconductors and (ii) P-type semiconductors

(i) N-type semiconductors — this type of semiconductor is obtained when a pentavalent material like antimony is added to pure semiconductor germanium crystal.
In N-type semiconductors, electrons are the majority carriers while holes constitute the minority carriers. Hence, N-type semiconductors conducts principally by electrons in the nearly empty conduction band and the process is called <u>excess conduction</u>

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(ii) P-type Extrinsic semiconductor

This type semiconductor is obtained when traces of a trivalent like boron (B) are added to a pure germanium crystal. In this case, the three valence electrons of boron atom forms covalent bonds with four surrounding atom but one bond is left incomplete and gives rise to a hole. Holes form the majority carriers while electrons constitute minority carriers.

~~BIPOLAR JUNCTION TRANSISTOR~~

BIPOLAR TRANSISTOR

Bipolar junction transistors (also called bipolar transistors) are a type of doped semiconductor material electronic device that contains three terminals. These are used in switching applications or amplification applications. Because they contain both holes and electrons, these components are named bipolar transistors.

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Bipolar (junction transistor) type of
bipolar (transistors) are a type of semi-
semiconductor material electronic device
that contains three terminals. These
are used in switching applications or
amplification applications. Because the
carrier both holes and electrons, the
components are named bipolar transistors.

The bipolar transistor basic construction (C)
consists of two PN-junctions producing
three connecting terminals with each
terminal being given a name to identify
it from the other two. These three
terminals are known and labelled as
Emitter(E), the Base(B) and the
Collector(C) respectively.

Bipolar Transistors are current regulating
devices that control the amount of
current flowing through them in
proportion to the amount of biasing voltage
applied to their base terminal acting
like a current controlled switch.

The principle of operation of the two
transistor types PNP and NPN is
exactly the same the only difference being
in their biasing and the polarity of the
power supply for each type.

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BIPOLAR TRANSISTOR

Bipolar junction transistors (bipolar transistors) are a semiconductor material device that contains three terminals are used in switching and amplification applications contain both holes and electrons components are named

1. Emitter

It is more heavily doped than any of the other regions because its main function is to supply majority charge carrier (either electrons or holes) to the base.

2. Base

It forms the middle section of the transistor. It is very thin as compared to either the emitter or collector and is very lightly doped

3. Collector

Its main function is to collect majority charge carriers coming from the emitter and passing through the base.

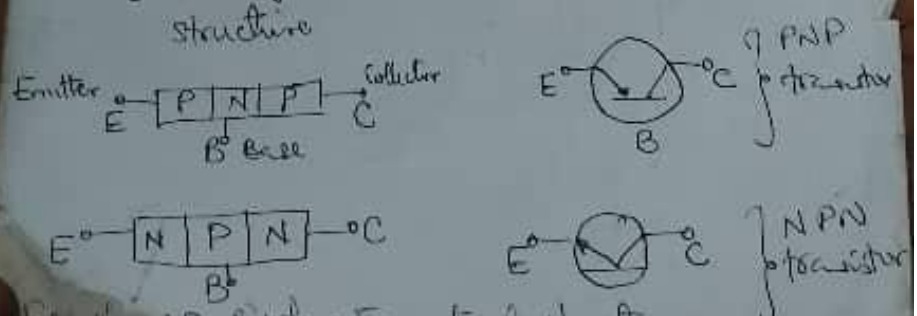

Fig: Bipolar Transistor Construction

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bound very rigidly to their parents atoms, thus requiring very large electric field to remove them from the attraction of their nuclei. In other words, insulators have no free charge carriers available with them under normal conditions.

In terms of energy bands, it means that insulators

1. have a full valence band,
2. have an empty conduction band,
3. have a large energy gap (of several eV) between them and
4. at ordinary temperatures, the probability of electrons from full valence band gaining sufficient energy so as to surmount energy gap and thus become available for conduction in the conduction band, is slight.

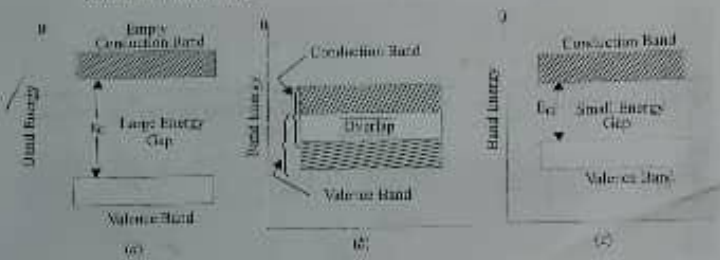

Fig. 01.14

This is shown in Fig. 51.14 (a). For conduction to take place, electrons must be given sufficient energy to jump from the valence band to the conduction band. Increase in temperature enables some electrons to go to the conduction band which fact accounts for the negative resistance-temperature coefficient of insulators.

Insulators

(ii) **Conductors.** Put in a simple way, conducting materials are those in which plenty of free electrons are available for electric conduction.

In terms of energy bands, it means that electrical conductors are those which have overlapping valence and conduction bands as shown in Fig. 51.14 (b).

In fact, there is no physical distinction ***between the two bands.*** Hence, the availability of a large number of conduction electrons.

Another point worth noting is that in the absence of forbidden energy gap in good conductors, there is no structure to establish holes. The ***total current in such conductors is simply a flow of electrons.*** It is exactly for this reason that the existence of holes was not discovered until semi conductors were studied thoroughly.

(iii) **Semiconductors.** A semiconductor material is one whose electrical properties lie in between those of insulators and conductors. Examples are : germanium and silicon.

In terms of energy bands, semiconductors can be defined as those materials which have almost an empty conduction band and almost filled valence band with a very narrow energy gap (of the order of 1 eV) separating the two.

At 0°K, there are no electrons in the conduction band and the valence band is completely filled. However, with increase in temperature, width of the forbidden energy bands is decreased so that

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some of the electrons are liberated into the conduction band. In other words, conductivity of semiconductor increases with temperature. Moreover, such departing electrons leave behind positive holes in the valence band (Fig. 51.12). Hence, semiconductor current is the sum of electron and hole currents flowing in opposite directions.

51.16. Crystal Structure

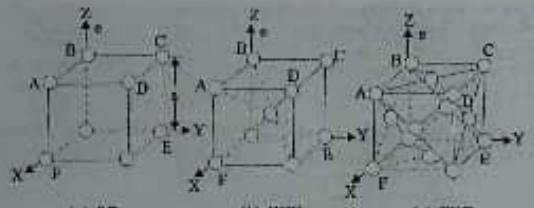
Semiconductors like germanium and silicon, have crystalline structure. That is the atoms are arranged in a three dimensional periodic fashion. The periodic arrangement of atoms in a crystal is called *lattice*. In a crystal, an atom stays far from a single, fixed position. The thermal vibrations associated with the atom are confined near this position. For a given semiconductor (silicon or germanium), there is

Cut one conductor through centre of diode

a unit cell that is representative of the entire lattice. By repeating the unit cell throughout the crystal, we can generate the entire lattice.

There are several different types of crystal lattice depending upon the symmetry and internal structure. One of them is the cubic crystal lattice. There are three basic types of unit cells in a cubic crystal lattice. These are simple cubic (SC), body centered cubic (BCC) and face centered cubic (FCC). Fig. 51.13 (a) shows a SC crystal. In this unit cell, each corner of the cubic lattice is occupied by an atom (indicated by a small sphere) that has six equidistant nearest neighbouring atoms. The dimension "a" is called the *lattice constant*. Only *polium* is crystallised into the simple cubic lattice.

Fig. 51.13 (b) shows a BCC unit cell. In this unit cell, each atom has eight nearest neighbouring atoms. Crystals exhibiting the BCC lattices include those of *sodium* and *tungsten*. Fig. 51.13 (c) shows a FCC unit cell. This unit cell has one atom at each of the six cubic faces in addition to the eight corner atoms. In an FCC lattice, each atom has 12 nearest neighbouring atoms. A large number of elements exhibit the FCC lattice form. These include *aluminium, copper, gold* and *platinum*.



(a) SC (b) BCC (c) FCC

Fig. 51.13

Fig. 51.14 shows the crystal structure of silicon and germanium—the elements semiconductors. This type of structure is called diamond crystal structure and it belongs to cubic crystal family. All atoms are identical in a diamond lattice. Note that each atom in diamond lattice is surrounded by four equidistant neighbours that lie in the corners of a tetrahedron.

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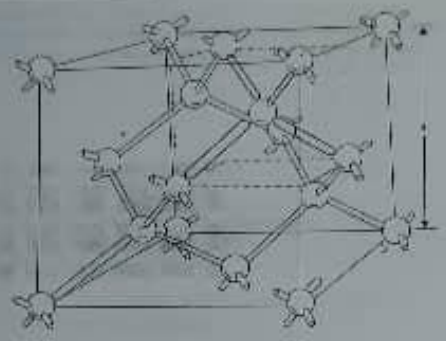

Fig. 51.16

51.17. Representation of Crystal Planes and Directions

Referring to Fig. 51.15 (a) again, we note that there are four atoms in the ABCD plane and five atoms in ACDF plane (four atoms from the corners and one from the centre). Moreover, the atomic spacing are different for the two planes. Therefore the crystal properties along different planes are different. The electrical and other device characteristics are dependent on the crystal orientation. A convenient method of defining or representing the various planes in a crystal is to use *Miller indices*. These indices are obtained using the following steps

1. Find the intercepts of the plane on the three coordinate axes in terms of lattice constant.
2. Take the reciprocals of these numbers and reduce them to the smallest three integers having the same ratio.
3. Enclose the result in parentheses (*hkl*) as the Miller indices for a single plane.

For example, consider the plane ABC having the intercept at a, $2a$ and $3a$ along the three rectangular coordinate axes as shown in Fig. 51.17. Taking the reciprocal of these intercepts, we get 1, 1/2, 1/3. Multiplying each fraction by the least common multiplier 2, we find that the smallest three integers having the same ratio are 2, 1 and 1. Thus the Miller indices from the plane are (211). The plane ABC can also be referred to as (211) plane.

Fig. 51.17

Fig. 51.18 shows the Miller indices of important planes in cubic crystal. It may be noted that for a plane that intercepts the X-axis on the negative side of the origin, the Miller indices are represented by ($\bar{h}$ k l). Similarly for a plane that intercepts the Y axis on the negative side of the origin, the Miller indices are represented by (h $\bar{k}$ l) and so on.

Sometimes it convenient to represent the Miller indices of the planes of equivalent symmetry by {hkl}. For example, in cubic crystal, the planes (100), (010), (001), ($\bar{1}$00), (0$\bar{1}$0) and (00$\bar{1}$) have the same symmetry. Therefore, these planes can be represented by Miller indices as {100}. Note that we have used the curly brackets to represent a set of planes with equivalent symmetry.

The line that originates from the origin and passes through the plane at right angles to it is called *crystal direction*. The crystal direction is indicated by enclosing the Miller indices in a square brackets

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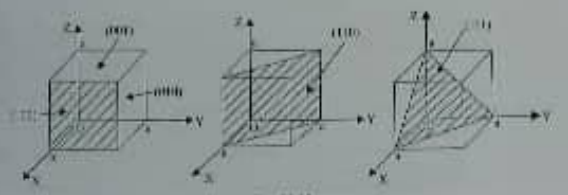

Fig. 51.18

i.e., [hkl]. For example, the direction illustrated by [100] is a direction for the X-axis and it is perpendicular to (100) plane. Similarly [111] direction is perpendicular to (111) plane. It is possible to represent a set of equivalent directions by < hkl >. Notice the use of curly signs. Thus < 100 > represents a set of directions for [100], [010], [100], [010] and [001].

51.16. Atomic Binding in Semiconductors

Bonds in semiconductors

Semiconductors like germanium* and silicon have crystalline structure. Their atoms are arranged in an ordered array known as crystal lattice. Both these materials are tetravalent i.e. each has four valence electrons in its outermost shell. The neighbouring atoms form covalent bonds by sharing four electrons with each other so as to achieve inert gas structure (i.e. 8 electrons in the outermost orbit). A two-dimensional view of the germanium crystal lattice is shown in Fig. 51.19 (b) in which circles represent atom cores consisting of the nuclei and inner 28 electrons. Each pair of lines represents a covalent bond. The dots represent the valence electrons. It is seen that each atom has 8 electrons under its influence.

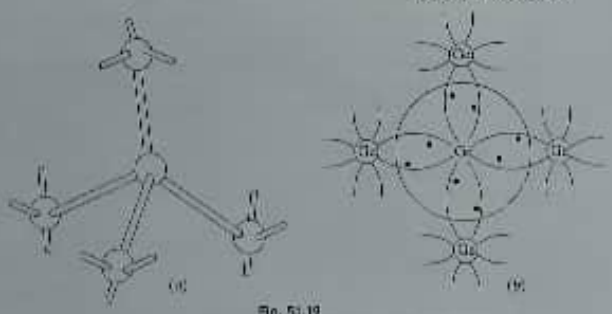

Fig. 51.19

* A single germanium atom has 32 electrons out of which only four electrons take part in electrical properties of germanium, the remaining 28 electrons being tightly bound to the nucleus. The four electrons revolving in the outermost orbit are called valence electrons.

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A 3-dimensional view of germanium crystal lattice is shown in Fig. 51.19 (a) where each atom is surrounded symmetrically by four other atoms forming a tetrahedral crystal. Each atom shares a valence electron with each of its four neighbours, thereby forming a stable structure.

In the case of pure (i.e. intrinsic) germanium, the covalent bonds have to be broken to provide electrons for conduction. There are many ways of rupturing the covalent bond and thereby setting the electrons free. One way is to increase the crystal temperature above 0°K.

It may be noted that covalent crystals are characterised by their hardness and brittleness. Their brittleness is due to the fact that in such crystals, adjacent atoms must remain in accurate alignment since the bond is strongly directional and forms along a line joining the atoms. The hardness is due to the great strength of the covalent bond itself.

51.19. Types of Semiconductors

Semiconductors can be classified as shown below :

Semiconductors
- intrinsic or pure semiconductors
- extrinsic or impure semiconductors
  - N-Type
  - P-Type

51.20. Intrinsic Semiconductors

An intrinsic semiconductor is one which is made of the semiconductor material in its extremely pure form.

Common examples of such semiconductors are : pure germanium and silicon which have forbidden energy gaps of 0.72 eV and 1.1 eV respectively. The energy gap is so small that even at ordinary room temperature, there are many electrons which possess sufficient energy to jump across the small energy gap between the valence and the conduction bands. However, it is worth noting that for each electron liberated into the conduction band, a positively charged hole is created in the valence band (Fig. 1.20). When an electric field is applied to an intrinsic semiconductor at a temperature greater than 0°K, conduction electrons move to the anode and the holes in the valence band move to the cathode. Hence semiconductor current consists of movement of electrons and holes in opposite directions. Electron current is due to movement of electrons in the conduction band whereas hole current is within the valence band as a result of the holes 'jumping' from one atom to another.

As stated above, in pure semiconductors, electric conduction is due to the thermally-generated electron hole pairs. Hence in pure semiconductors kept in the dark, thermally generated charge carriers are the only means of conduction. The number of such charge carriers per unit volume (i.e. intrinsic carrier density) is given by

$$n_i = N \exp(-E_g/2kT)$$

where N is constant for a given semiconductor, E_g is the band gap energy in joules, k is Boltzmann's constant and T is the temperature in °K.

Example 51.2. *Find the intrinsic carrier concentration in silicon at 250° K for which $N = 3 \times 10^{25}$ m^{-3}, $E_g = 1.1$ eV, $\mu_p = 0.14$ m^2/V-s and $\mu_n = 0.05$ m^2/V-s. Also, find the conductivity of silicon.* (Electronics-II, Madras Univ. 1993)

Solution. The intrinsic carrier concentration in pure silicon is given by

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$$n_i = N \exp(-E_g/2kT)$$

Now, $N = 3 \times 10^{29}\ m^{-3}$, $E_g = 1.1\ eV = 1.1 \times 1.6 \times 10^{-19} = 1.76 \times 10^{-19}\ J$

$k = 1.38 \times 10^{-23}\ J/K$, $T = 300°K$

$\therefore$ $n_i = 3 \times 10^{29} (-1.76 \times 10^{-19}/2 \times 1.38 \times 10^{-23} \times 300) = \mathbf{1.76 \times 10^{16}\ m^{-3}}$

$\sigma = n_i e (\mu_e + \mu_h) = 1.76 \times 10^{16} \times 1.6 \times 10^{-19} (0.14 + 0.05) = \mathbf{0.535 \times 10^{-5}\ S/m}$

51.21. Hole Formation in Semiconductors

The formation of a hole which is a positive charge carrier is explained below :

As shown in Fig. 51.21, suppose the covalent bond is broken at *A* and the electron has moved through the crystal lattice leaving behind a hole in the covalent bond. An electron at *B* may jump into the vacant hole at *A* and later, an electron at *C* may jump into the hole at *B* and so on. In this way, by a succession of electron movements, a hole will appear at *G* and a negative charge would have moved from *G* to *A*. It would, however, be more convenient to regard positive charge to have moved from *A* to *G* and this conception gives rise to a hole as a positive charge carrier as if it were an electron with a positive charge. It should be clearly understood that these holes are due to the movement of electrons in the valence band and that each electron movement corresponds to a collision. The drift velocity of holes is, obviously, much less than the drift velocity of electrons.

Alternatively, an intrinsic semiconductor may be defined as one ***in which the number of conduction electrons is equal to the number of holes.***

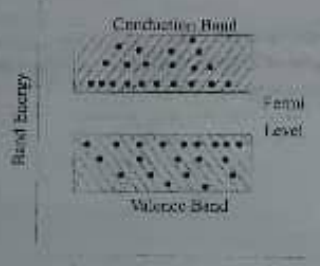

Fig. 51.20

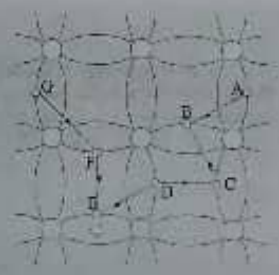

Fig. 51.21

Schematic energy band diagram of an intrinsic semiconductor at room temperature is shown in Fig. 51.20. Only two bands *i.e.* valence and conduction bands have been shown since lower filled bands are not of any consequence. Here, Fermi level* lies exactly in the middle of the forbidden energy gap.

51.22. Fermi Level in an Intrinsic Semiconductor

It can be proved that in an intrinsic semiconductor, Fermi energy level* E_F lies in the middle of the energy gap *i.e.* midway between the conduction and valence bands.

Let, at any temperature *T*°K

n_c = No. of electrons in the conduction band

n_v = No. of electrons in the valence band

* For the present discussion, Fermi level may be defined as the energy which corresponds to the centre of gravity of conduction electrons and holes weighted according to their energies.

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$$N = n_c + n_v$$

= No. of electrons in both bands.

Further, let us make the following simplifying assumptions :

1. width of energy bands are small as compared to forbidden energy gap between them;
2. since band widths are small, all levels in a band have the same energy;
3. energies of all levels in valence band are zero as shown in Fig. 51.22;
4. energies of all levels in the conduction band are equal to E_c

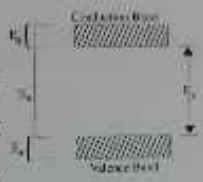

Fig. 51.22

In Fig. 51.22, the zero-energy reference level has been arbitrarily taken at the top of valence band.

Now, number of electrons in the conduction band is

$$n_c = N P(E_c)$$

where $P(E_c)$ represents the probability of an electron having energy E_c. Its value can be found from Fermi-Dirac probability distribution function given by

$$P(E) = \frac{1}{1 + e^{(E - E_F)/kT}}$$

where $P(E)$ is the probability of finding an electron having any particular value of energy E and E_F is Fermi level.

$$\therefore \quad P(E_c) = \frac{1}{1 + e^{(E_c - E_F)/kT}} \qquad \therefore \quad n_c = \frac{N}{1 + e^{(E_c - E_F)/kT}}$$

Now, number of electrons in the valence band is $n_v = N P(0)$

The probability $P(0)$ of an electron being found in the valence band with zero energy can again be calculated by putting $E = 0$ in the Fermi-Dirac probability distribution function.

$$P(0) = \frac{1}{1 + e^{(0 - E_F)/kT}} = \frac{1}{1 + e^{-E_F/kT}}. \qquad \text{Hence, } n_v = \frac{N}{1 + e^{-E_F/kT}}$$

Now $$N = n_c + n_v \quad \text{or} \quad N = \frac{N}{1 + e^{(E_c - E_F)/kT}} + \frac{N}{1 + e^{-E_F/kT}}$$

$$\therefore \quad 1 = \frac{1}{1 + e^{-E_F/kT}} = \frac{1}{1 + e^{(E_c - E_F)/kT}}$$

which on simplification gives $E_F = E_c/2$.

Hence, it shows that in an intrinsic semiconductor, the Fermi level lies midway between the conduction and valence bands. That this conclusion is physically possible can be best seen with the help of Fig. 51.23.

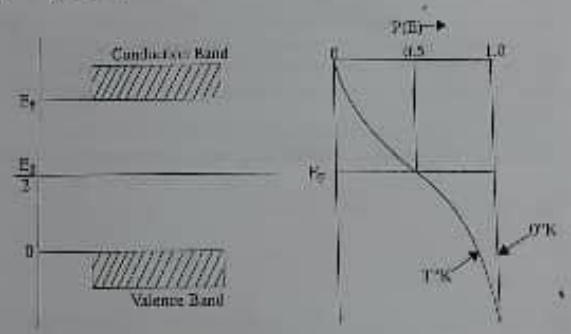

Fig. 51.23

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51.23. Extrinsic Semiconductors

Those intrinsic semiconductors to which some suitable impurity or doping agent or dopant has been added in extremely small amounts (about 1 part in 10^8) are called *extrinsic* or impurity semiconductors.

The usual doping agents are :

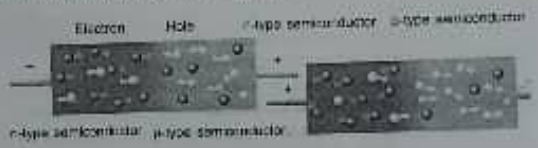
1. ***pentavalent*** atoms having five valence electrons (arsenic, antimony, phosphorus) or
2. ***trivalent*** atoms having three valence electrons (gallium, indium, aluminium, boron).

Pentavalent doping atom is known as ***donor*** atom because it donates or contributes one electron to the conduction band of pure germanium. The trivalent atom, on the other hand, is called ***acceptor*** atom because it accepts one electron from the germanium atom.

Depending on the type of doping material used, extrinsic semiconductors can be sub-divided into two classes :

(i) *N*-type semiconductors and (ii) *P*-type semiconductors.

(*a*) ***N*-type Extrinsic Semiconductor.** This type of semiconductor is obtained when a pentavalent material like antimony (Sb) is added to pure germanium crystal. As shown in Fig. 51.24 (*a*), each antimony atom forms covalent bonds with the surrounding four germanium atoms with the help of four of its five electrons. The fifth electron is superfluous and is loosely bound to the antimony atom. Hence, it can be easily excited from the valence band to the conduction band by the application of electric field or increase in thermal energy. Thus, practically every antimony atom introduced into the germanium lattice, contributes one conduction electron into the germanium lattice ***without creating a positive hole***. Antimony is called *donor* impurity and makes the pure germanium an *N*-type (*N* for negative) extrinsic semi-conductor. As an aid to memory, the student should associate the N in deNor with N in the *N*-type material and *N* in Negative charge carrier.



It may be noted that by giving away its one valence electron, the donor atom becomes a positively-charged *ion*. But it cannot take part in conduction because it is firmly fixed or tied into the crystal lattice. It will be seen that apart from electrons and holes ***intrinsically available in germanium***, the addition of antimony greatly increases the number of conduction electrons. Hence, concentration of electrons in the conduction band is increased and exceeds the concentration of holes in the valence band. Because of this, Fermi level shifts upwards towards the bottom of the conduction band as shown in Fig. 51.24 (*a*),* because the number of charge carriers has become more in conduction band than in valence band.

In terms of energy levels, the fifth antimony electron has an energy level (called donor level) just below the conduction band. Usually, the donor level is 0.01 eV below conduction band for germanium and 0.054 eV for silicon.

It is seen from the above description that in *N*-type semiconductors, electrons ***are the majority carriers*** while holes constitute the minority carriers. Hence, *N*-type semiconductor conducts principally by electrons in the nearly empty conduction band and the process is called '*excess*' conduction.

Another point worth noting is that even though *N*-type semiconductor has excess of electrons,

* Since the number of electrons as compared to the number of holes increases with temperature, the position of Fermi level also changes considerably with temperature.

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